

Rate equations

A summary of rate equations useful for modeling, organized by application. Sources are given in parentheses.

H. Heat transfer

H1. Fourier's law - conduction (Incropera, 2003)

$$q''_x = -k \cdot \frac{dT}{dx}$$

q''_x is heat flux e.g. W m^{-2}

k is thermal conductivity e.g. $\text{W m}^{-1} \text{K}^{-1}$

T is temperature of course, $^{\circ}\text{C}$ or K

$$q''_x = k \cdot \Delta T / \Delta x$$

Plane wall, steady state

H2. Newton's law of cooling - convection (Incropera, 2003)

$$q'' = h(T_s - T_{\infty})$$

q'' is heat flux from surface

h is the (convection) heat transfer coefficient

M. Mass transfer

M1. Fick's law - diffusion (Incropera, 2003)

$$j''_x = D \cdot \frac{dc}{dx}$$

j''_x = Mass flux, e.g. $\text{g m}^{-2} \text{s}^{-1}$

D is diffusivity e.g. $\text{m}^2 \text{s}^{-1}$

c is concentration e.g. g m^{-3}

$$j''_x = D \cdot \frac{\Delta c}{\Delta x}$$

Plane wall at steady state.

M2. Mass transfer coefficient approach - convection or diffusion (Cussler, 2009)

$$j'' = k \cdot \Delta c$$

k is mass transfer coefficient, e.g. $\text{m}^2 \text{s}^{-1}$

X. Microbial activity

X1. Monod equation - substrate utilization rate (Rittmann and McCarty, 2001)

$$r_{ut} = \frac{\hat{q} \cdot S}{K + S} \cdot X_a$$

r_{ut} is substrate utilization rate, e.g. $g\ m^{-3}\ s^{-1}$.

$\hat{q}$ is maximum (specific) utilization rate, e.g.

$$g\ g_s^{-1}\ s^{-1}.$$

S is substrate concentration, e.g. $g\ m^{-3}$ (usually COD).

K is S concentration that gives $1/2$ maximum rate

X_a is active microbial biomass concentration, e.g. $g\ m^{-3}$ (usually COD).